SOP-YEL-ACC-67: Resolving Huawei Access Node QoS Congestion Event

Node: YEL-ACC-67 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_4553

Description

This Standard Operating Procedure outlines the step-by-step approach to diagnose and resolve QoS congestion events (microbursts, interface tail drops, QoS queue starvation) on the Huawei Access Node YEL-ACC-67 at the Yelahanka Sub-station. The congestion event is triggered by the QoS policing function, which monitors the traffic rate and discards excess packets to prevent network congestion. However, in this specific scenario, the policing function is not functioning correctly, resulting in the accumulation of packets and the subsequent congestion event. This is likely due to a misconfiguration of the QoS policing profile or a hardware issue with the interface. The root cause analysis will involve verifying the configuration of the QoS policing profile, checking the interface hardware, and investigating the traffic patterns to determine the source of the congestion event.

Pre-requisite Checks

- Verify that the remote hands have been dispatched to the Yelahanka Sub-station to perform physical checks on the node.

- Check the status of the redundant link to ensure that it is functioning correctly and can be used as a fallback if necessary.

- Verify that the node is operating in the correct mode and that the QoS policing function is enabled.

Potential Root Causes

Cause	Probability
Misconfigured QoS policing profile	High
Hardware issue with the interface	Medium
Unusual traffic patterns	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policer, Expected Output: QoS policing profile configuration, Description: This command will display the configuration of the QoS policing profile, including the policing rate, burst size, and action taken when the policing threshold is exceeded. This will help verify that the policing function is configured correctly.

Command: display interface, Expected Output: Interface status and statistics, Description: This command will display the status and statistics of the interface, including the current bandwidth utilization, packet drops, and errors. This will help identify any hardware issues with the interface.
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```
Command: display traffic, Expected Output: Traffic patterns and statistics, Description: This command will display the traffic patterns and statistics, including the source and destination IP addresses, packet sizes, and protocols. This will help identify any unusual traffic patterns that may be contributing to the congestion event.
```

Resolution Steps

Step 1: Step: Verify QoS policing profile configuration, Command: display qos policer, Description: Verify that the QoS policing profile is configured correctly and that the policing rate and burst size are set to the desired values. If the configuration is incorrect, update the configuration to the correct values.

Step 2: Step: Update interface hardware, Command: upgrade interface, Description: If the issue is with the interface hardware, update the interface hardware to the latest revision. This may involve replacing the interface card or updating the firmware.

Step 3: Step: Adjust QoS policing profile, Command: modify qos policer, Description: If the issue is with the QoS policing profile, adjust the policing rate and burst size to the desired values. This may involve increasing the policing rate or decreasing the burst size to prevent packet drops.

Post-Check Verification

```
Command: display qos policer, Expected Output: QoS policing profile configuration, Description: Verify that the QoS policing profile is configured correctly and that the policing rate and burst size are set to the desired values.

Command: display interface, Expected Output: Interface status and statistics, Description: Verify that the interface is functioning correctly and that packet drops and errors have been minimized.

Command: display traffic, Expected Output: Traffic patterns and statistics, Description: Verify that the traffic patterns are normal and that there are no unusual traffic patterns contributing to the congestion event.
```

Roll-back Procedures

- Step: Revert QoS policing profile configuration, Command: undo modify qos policer, Description: If the fix fails, revert the QoS policing profile configuration to the previous values.

- Step: Revert interface hardware, Command: undo upgrade interface, Description: If the fix fails, revert the interface hardware to the previous revision.

- Step: Revert QoS policing profile, Command: undo modify qos policer, Description: If the fix fails, revert the QoS policing profile to the previous values.